

Woocheol Shin

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Research Interest

Education

KAIST

M.S. Graduate School of Data Science

Feb. 2025 ~ Present

UNIST

B.S. Computer Science and Engineering

Feb. 2019 ~ Feb. 2025

Research Experience

System Intelligence Lab, KAIST

Undergraduate Research Intern

Advisor : Prof. Jinkyoo Park(jinkyoo.park@kaist.ac.kr)

Oct. 2024 ~ Feb. 2025

Data Mining Lab, UNIST

Undergraduate Researcher, Seminar Organizer

Advisor : Prof. Junghoon Kim (junghoon.kim@unist.ac.kr)

Focus : Network stability, Cohesive subgraph discovery

Dec. 2022 ~ Jun. 2024

Research Papers

Diffusion Fine-Tuning via Reparameterized Policy Gradient of the Soft Q-Function

Hyeongyu Kang*, Jaewoo Lee*, Woocheol Shin*, Kiyoungh Om, Jinkyoo Park

ICLR 2026

Diffusion Alignment as Variational Expectation-Maximization

Jaewoo Lee, Minjae Kim, Seungyong Choi, Inseop Song, Seongyun Yun, Hyeongyu Kang, Woocheol Shin,

Taegon Yun, Kiyoungh Om, Jinkyoo Park

ICLR 2026

Urban Traffic Network Layout Optimization with Guided Discrete Diffusion Models

Taegon Yun, Inseop Song, Woocheol Shin, Yejin Shin, Seunghyun Woo, Sungsu Lim, Jinkyoo Park

WSDM 2026

Beyond trivial edges: A fractional approach to cohesive subgraph detection in hypergraphs

Hyewon Kim, Woocheol Shin, Dahee Kim, Junghoon Kim*, Sungsu Lim, Hyunji Jeong

Knowledge-Based Systems 2025

Complex Network Stability: Exploring Comprehensive Anchored Edge s-Core

Taejoon Han*, Hyewon Kim*, Seongsik Hwang, Woocheol Shin, Junghoon Kim*, Jaehyun Bhang, Hyeonggi Hong

Korea Database Conference (WiP track) 2024

Awards and Achievements

ISE & GSDS Research Day Poster Competition, KAIST

Excellence Award

For the poster *Diffusion Finetuning via Reparameterized Policy Gradient of the Soft Q-function*

Nov. 2025

Dean's List, UNIST CSE

Awarded to top 4 students out of 224

Sep. 2024

Superior semester grades, UNIST

Spring 2023 (4.3/4.3), Fall 2023 (4.24/4.3), Spring 2024 (4.3/4.3)

Scholarship

- KAIST Scholarship, KAIST Whole semesters
- Academic Performance Scholarship, UNIST Whole semesters

Patent

- Data Analysis System and Method for Detecting Cohesive Subgraph in Hypergraphs Korean Patent Application, Aug. 2025
 - Application No. 10-2025-0107601
 - Applicants: UNIST; Industry-Academic Cooperation Foundation, Kongju National University
 - Inventors: Junghoon Kim, Hyewon Kim, Woocheol Shin, Dahee Kim, Hyunji Jeong

Teaching Experience

- Teaching Assistant, KAIST Sep. 2025 – Dec. 2025
 - IE437: Data-Driven Decision Making and Control

Project

- AICP, UNIST Apr. 2023 ~ Dec. 2023
 - UNIST AI Challengers Program
 - Topic : Bipartite Cohesive Subgraph Discovery in Networks
- Creative Independent Project, UNIST AIGS Oct. 2023 ~ May. 2024
 - Topic : A Budget Minimisation to Reinforcing Social Engagement
- UIRP, UNIST Mar. 2024 ~ May. 2024
 - UNIST Undergraduate Interdisciplinary Research Project
 - Topic : Generalised Anchoring Methods for Maximising User Engagement in Complex Social Networks

Industrial Projects

- Collaborative Project with ETRI Feb. 2025 ~ Present
 - Topic : Reinforcement Learning-based Job Scheduling Policy Learning

Other Experiences

- NAVER CONNECT 1784 School Revolution Program, NAVER Dec. 2022 ~ Feb. 2023
 - Teaching AI, SW, and Autonomous Vehicles
- ARAMCO Coding school, ARAMCO Aug. 2022 ~ Nov. 2022
 - Teaching Programming, IOS app development
- Education donation activities by university students, KOFAC May. 2019 ~ Jul. 2019
 - Teaching Programming

References

- Assistant Professor, Junghoon Kim
 - Computer Science and Engineering
 - UNIST
 - Email: junghoon.kim@unist.ac.kr
- Associate Professor, Jinkyoo Park
 - Department of Industrial and Systems Engineering
 - Associate Professor (Adjunct), Graduate School of Data Science
 - Associate Professor (Adjunct), Graduate School of AI
 - KAIST
 - Email: jinkyoo.park@kaist.ac.kr

